

Shuai Ma

Email: mashuai@gmail.com | Web: shuaima-research.github.io | Shanghai, China

Education

Seattle University

Seattle, WA, United States

Albers School of Business and Economics

M.S. in Business Analytics, STEM-designated

2018–2020

- Relevant coursework: programming for business analytics, applied econometrics, statistical learning, forecasting, big-data analytics, data management, data visualization, and data architecture.
- Capstone: *Influential Factors on Residential Property Prices Using Data-Mining Methods*; analyzed Pierce County housing transactions using statistical and machine-learning method to predict residential prices.

University of Cincinnati

Cincinnati, OH, United States

College of Design, Architecture, Art, and Planning

M.S. in Architecture, research-based program; research and thesis focused on urban development

2009–2011

- Research training: urban theory, research methods, pre-thesis research, social-science research for planners, land-use control, international development policy, and thesis-based scholarly writing.
- Thesis: *Impact of High-Speed Rail Station Development on Surrounding Urban Form: The Beijing–Tianjin Case*.

Hebei University of Technology

Tianjin, China

Project 211 and Double First-Class university

B.Eng. in Urban Planning and Design

2003–2008

- Undergraduate planning foundation: planning studios, regional planning, urban design, urban economics, urban geography, transportation planning, urban ecology, planning code, and spatial information technology.
- NACES course-by-course evaluation: equivalent to a U.S. bachelor's degree in Urban and Regional Planning.

Research Interests

Substantive: housing and urban policy; climate adaptation and flood mitigation; place-based public investment; land-market capitalization; affordability, neighborhood sorting, and distributional effects.

Methods: causal inference and quasi-experimental policy evaluation; spatial econometrics; GIS and urban data science.

Research Papers and Writing Samples

Hazard or Amenity? Flood Risk, Waterway Proximity, and Neighborhood Inequality

2025–Present

Sole-authored PhD writing sample; empirical study of Pierce County, Washington

- Examines how flood risk, waterway proximity, and neighborhood inequality are jointly reflected in residential prices, using 47,763 single-family sales (2021–2025) linked to parcel, FEMA flood-risk, and neighborhood data; recasts whether flood risk is discounted into how hazard, amenity, and neighborhood advantage occupy the same ground.
- Combines hedonic regression (tract-clustered standard errors) with spatial analysis (Moran's I, LISA), a flood-income–waterway typology, and machine-learning checks, finding no reliable flood-risk discount because hazard and waterfront amenity are too co-located for price alone to separate.
- Shows that where hazard and amenity share the same parcels, prices cannot reveal who bears flood risk, so value-oriented flood-mitigation investment tends to flow to already-advantaged neighborhoods.

Strategic Management's Foundational Silence

2025–Present

First-author manuscript in revision, coauthored with Rubina Mahsud and Greg Prussia; developed from a 2020 Strategic Management Society conference proposal

- Built and cleaned a corpus of 3,202 Strategic Management Journal articles and applied classical natural language processing methods, local open-source large language models, and cloud-based models to analyze the field's dominant vocabulary.
- Develops the concept of “foundational silence” to argue that poverty, war, disease, hunger, homelessness, and related human concerns should be treated as central social contexts rather than peripheral externalities.

Conference Presentations

Strategic Management Society 2060: From Effectiveness to Impact, from Success to Significance

October 2020

Strategic Management Society Annual Conference, Virtual

Mahsud, R., and Ma, S. Presented by Shuai Ma. Conference proposal, later developed into a first-authored manuscript.

Selected Research and Analytical Projects

Pro Forma Urbanism: Financial Feasibility Modeling and Land Value Capture in Shenzhen Urban Renewal 2025

Sole-authored research brief

- Reverse-engineered a state-owned developer's 45-slide feasibility deck, rebuilding its static and dynamic discounted-cash-flow pro forma as reproducible spreadsheets and validating them against the source's reported outputs (IRR 27.45 percent; NPV ≈ ¥2.44 billion), to show how the model prices land and decides what gets built.
- Reconstructed Shenzhen's FAR-graduated, tenure-contingent land-pricing rules to trace how density entitlements convert into a payable land premium, finding that roughly 86 percent of the premium falls on the new, market-priced density the plan creates—floor-area ratio functioning as a value-capture instrument.

Empirical Analysis of Guangzhou Residential Rental Markets 2021

Sole-authored hedonic study

- Collected and cleaned 3,000 Guangzhou rental listings and built a hedonic log-rent model with robust standard errors and multicollinearity diagnostics, finding floor area to be the dominant structural correlate of rent (elasticity ≈ 0.72).
- Estimated conditional district rent differentials to characterize the spatial structure of the rental market, finding that central, employment- and transit-rich districts retain substantial premiums after dwelling attributes are controlled (approximately 51 percent in Tianhe, 42 percent in Yuexiu, and 27 percent in Haizhu relative to Baiyun), and interpreting these as capitalization patterns rather than causal effects given the data's measurement limits.

Operationalizing Highest and Best Use: A GDV-Oriented Integer-Programming Approach to Product-Mix Optimization for Business-Park Development 2021

Initiated and led an internal real-estate analytics study

- Operationalized Highest and Best Use as an integer linear program that optimizes a business park's product mix under floor-area-ratio and plot-coverage constraints, solved to optimality in seconds with open-source tools.
- Used sensitivity analysis to quantify the GDV cost of prioritizing market absorption over pure value maximization, turning a single-objective optimum into a risk-aware product-mix recommendation.

Engineering Controlled Flexibility: New-type Industrial Land (M0) and the Governance of Innovation Space 2021

Sole-authored comparative policy analysis

- Coded industrial land-use policy across seven Pearl River Delta cities along six regulatory dimensions to theorize Shenzhen's M0 land category as "controlled flexibility" enforced through conditional property rights.
- Triangulated policy texts with 2021 market evidence to show how flexible industrial zoning drifts toward saleable office-like product, requiring continual loophole-closing and a spatial block-line counterweight.

Land Use as a Hydrological Variable: A Site-Conditioned Framework for Low-Impact Development 2011–2017

Sole-authored retrospective multi-case study

- Synthesized six sponge-city/LID commissions from watershed to CBD-plot scale into a comparative case series, arguing that land use is a first-order determinant of stormwater design and deriving five site-conditioned propositions linking terrain, soil, groundwater, and density to treatment-train configuration.
- Conducted the underlying hydrological modeling, including SWMM simulation of lake storage and ponding risk under design storms and runoff calculations showing land conversion doubling peak flow on a low-permeability site, to ground a land-use-conditioned LID design framework.

Auditing Carbon at the Scale of the Urban System: A Spatial Planning Emissions Model for Jiangsu Province 2011

Core project-team member

- Co-developed a multi-sector carbon accounting model integrating activity data across land use, buildings, industry, transportation, water, waste, and ecological space, and produced baseline and policy-scenario emission estimates (2005–2030) using per-capita and per-GDP intensity indicators for a 13-city provincial urban system.

Technical Skills

Econometric and Quantitative Methods: Applied econometrics (hedonic price modeling; OLS with clustered standard errors; model selection and multicollinearity diagnostics); statistical learning and forecasting; scenario and baseline modeling

Spatial and Data Methods: GIS and spatial analysis; spatial data integration and geoprocessing; integration and cleaning of large-scale, multi-source and multimodal data (spatial, tabular, and text); data visualization

Programming and Software: Python, R, SQL; GeoPandas, PuLP; ArcGIS/QGIS, SWMM; Quarto, Git/GitHub; Tableau, Excel, Adobe Illustrator

Professional Experience

Founder & Principal Researcher, Beamlinked Consulting Co., Ltd.

2023–Present

- Founded an independent research and advisory practice focused on housing and real estate markets, land-use policy, and urban development, with project engagements in Shandong, Guangzhou, Chongqing, and other regions across China.
- Conducted market, policy, and spatial analyses for housing, business-park, and urban-development projects, integrating administrative information, publicly available market data, land-use controls, and GIS to assess market demand, development feasibility, and product positioning.
- Developed quantitative and spatial workflows using Python, R, GIS, and administrative and publicly available data to support research and advisory assignments.

Vice Director, Property Asset Management, Ping An Group

2020–2023

Planning, research, and investment analysis lead within a real estate business unit

- Led housing and real estate market research and investment analysis for a business unit, producing recurring analytical reports for senior management on housing markets, urban policy, infrastructure investment, and industrial-park planning.
- Initiated quantitative studies on product positioning and highest-and-best-use optimization, evaluating how regulatory and market constraints affected product mix and gross development value.
- Conducted feasibility, market, and investment analysis for five industrial-park development projects with a combined planned investment of approximately RMB 3.75 billion, two of which are now operational.

Director, Planning and Development, China Fortune Land Development

2017–2018

- Directed planning and design across two public-private-partnership new-town districts (~3,200 acres each), spanning land use, urban design, road and infrastructure networks, flood-control and water systems, residential neighborhoods, parks and public facilities, within development contexts involving land acquisition and resident resettlement.
- Coordinated large-scale development across public and private actors, integrating land-use regulation, infrastructure provision, and real estate development.
- Co-developed a group-wide development guideline adopted as a standard for new real estate projects.

Project Manager, AtkinsRéalis / Atkins

2013–2017

- Led planning and design teams across town and master planning, business and industrial parks, commercial complexes, and urban development projects.
- Directed applied research on environmental and performance-based analysis (microclimate, energy, and building-physics simulation), applying quantitative modeling to development and design decisions.
- Advised project teams on green-building and sustainability standards (LEED, BREEAM, WELL, and equivalents).

Urban Planner, Arup

2011–2013

- Contributed to urban regeneration and redevelopment projects, including the Shanghai Expo Urban Best Practices Area (UBPA) and Knowledge & Innovation Community (KIC) renewal, alongside waterfront, transit-oriented, and regional planning with spatial and GIS-based analysis.
- Contributed to the Jiangsu Provincial Low-Carbon Urban System Strategy, supporting carbon-audit modeling and province-scale scenario analysis.
- Supported a National Basic Research Program study and contributed research materials incorporated into two published department-led books: *Low-Carbon Ecological Space: Rethinking Cross-Dimensional Planning Methods* and *Study on the Economics of Green Buildings in China*.

Assistant Planner, Part-Time, Tongji University Urban Planning and Design Institute

2007–2009

- Supported regional planning, comprehensive urban development planning, tourism planning, historic-district design, and industrial-park planning projects.

Certifications and Professional Credentials

LEED AP Neighborhood Development, U.S. Green Building Council

Honors and Awards

- Dean's Honor List, Albers School of Business and Economics, Seattle University, Fall 2018.
- Best Graduate Design Award, Hebei University of Technology, 2008.
- Second Prize, National Undergraduate Urban Design Competition, Urban Planning Discipline Steering Committee, Ministry of Construction, China, 2007.